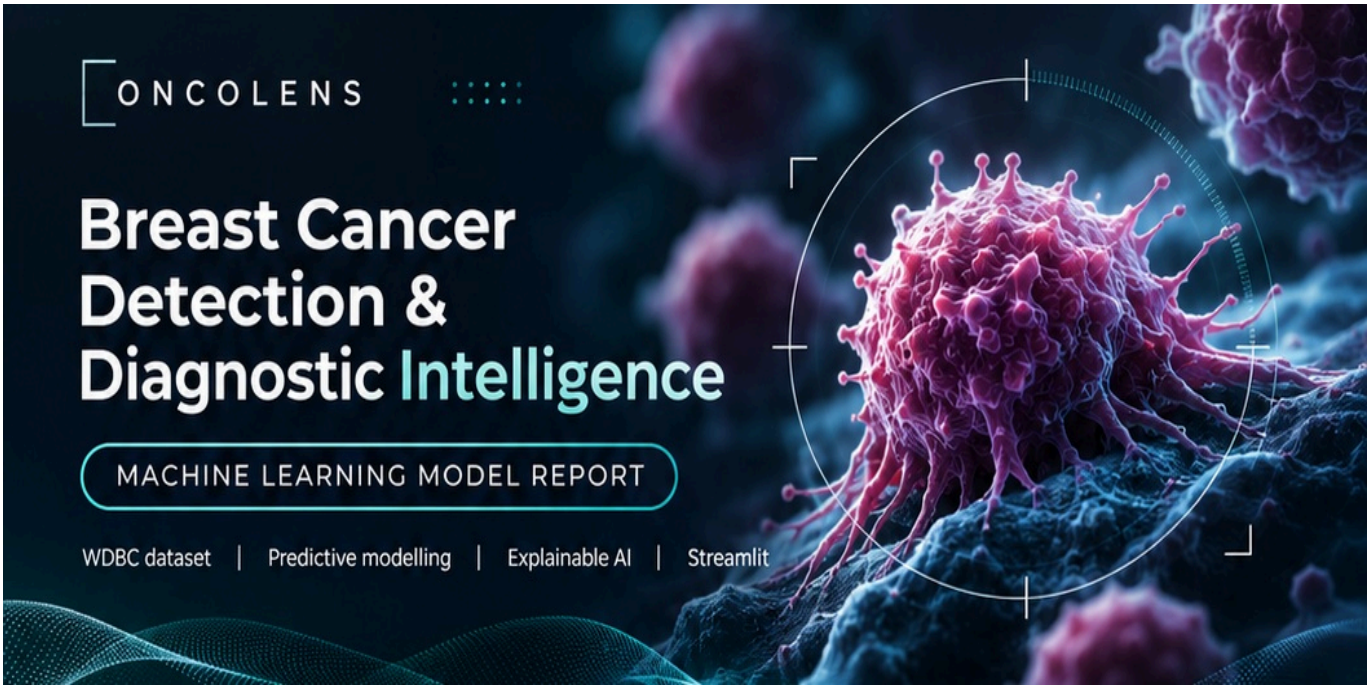


GITHUB REPOSITORY

github.com/nidhi-yadav20799/Breast-Cancer-Detection-ML

LIVE APPLICATION

oncolens-breast-diagnostic.streamlit.app

MACHINE LEARNING MODEL REPORT | WDBC DATASET | EXPLAINABLE AI

Educational research prototype · not for clinical diagnosis or medical decision-making.

1. Executive Summary

ONCOLENS is an educational machine-learning project that classifies Wisconsin Diagnostic Breast Cancer (WDBC) observations as benign or malignant using 30 numerical measurements derived from digitised fine-needle-aspirate (FNA) images. The workflow includes data validation, exploratory analysis, model comparison, hyperparameter tuning, feature-importance analysis and a deployed Streamlit prototype.

The saved estimator is a scikit-learn pipeline containing StandardScaler followed by Logistic Regression. It uses a 0.50 classification threshold. The application includes overview, single-sample analysis, batch CSV analysis, explainability and methodology workspaces.

IMPORTANT: ONCOLENS is a research and educational prototype, not a clinically validated diagnostic system. Its predictions must not be used for screening, diagnosis, treatment or medical decision-making.

2. Dataset and Project Requirements

The WDBC dataset contains 569 observations, 30 continuous predictors and two diagnosis classes. The target mapping is benign (B) = 0 and malignant (M) = 1. The predictors describe radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry and fractal dimension, each represented by mean, standard error and worst measurements.

Dataset attribute	Value
Observations / predictors	569 / 30
Benign / malignant	357 (62.74%) / 212 (37.26%)
Train / holdout split	455 / 114; stratified 80/20; random_state=42
Missing values / duplicate rows	0 / 0 in validated source
Input structure	30 numerical features; diagnosis target B/M

Implemented brief requirements include data cleaning and validation, exploratory data analysis, feature preprocessing, at least two ML algorithms, hyperparameter tuning, classification metrics, false-negative analysis and an interactive application. Five classifiers were evaluated: Logistic Regression, RBF-SVM, KNN, Random Forest and Gaussian Naive Bayes.

3. Data Preparation and EDA

The validation script checks the expected 569 × 32 schema, required columns, diagnosis labels, unique IDs, duplicate observations, numeric conversion, finite predictor values and non-negative measurements. A completely empty source column is removed, and the cleaned dataset is saved separately.

EDA covers class distribution, feature distributions, class-wise comparisons, Spearman correlations and IQR-based outlier screening. The README records strong correlations among radius, perimeter and area measurements, indicating redundancy among some geometric predictors. Potential outliers are investigated rather than automatically removed.

4. Modelling and Model Selection

Scale-sensitive estimators use StandardScaler within pipelines; Random Forest is evaluated without standard scaling. Stratified five-fold cross-validation is used. GridSearchCV tunes Logistic Regression and SVM on the training partition, using ROC-AUC for refitting.

Candidate	Best CV ROC-AUC	Recorded tuning / status
Logistic Regression	0.9959	C=1; L1; liblinear; class_weight=None
SVM (RBF)	0.9960	C=100; gamma=0.001; class_weight=None
KNN	Evaluated	Compared in model-development notebook
Random Forest	Evaluated	Compared in model-development notebook
Gaussian Naive Bayes	Evaluated	Compared in model-development notebook

The saved model is Logistic Regression. Its cross-validation ROC-AUC is effectively equivalent to SVM. In the recorded holdout split, Logistic Regression has higher malignant recall and two false negatives, compared with three for SVM.

5. Recorded Model Performance

The metrics below are recorded for the saved Logistic Regression model at threshold 0.50. Malignant is the positive class.

Metric	Recorded score
Accuracy	97.37%
Precision (malignant)	97.56%
Recall / sensitivity (malignant)	95.24%
Specificity	98.61%
F1-score	96.39%
ROC-AUC	0.9964

Confusion matrix (actual rows; predicted columns):

	Predicted benign	Predicted malignant
Actual benign	71 (TN)	1 (FP)
Actual malignant	2 (FN)	40 (TP)

The model correctly classified 40 of 42 malignant observations in this split. Two malignant observations were classified as benign (false negatives). Malignant recall and false-negative counts are therefore important alongside overall accuracy.

6. Explainability and Streamlit Application

The repository contains Logistic Regression coefficient analysis and permutation feature-importance outputs. These provide model-level insight into predictor influence and sensitivity to feature permutation; they do not establish causation or explain a clinical outcome.

Application workspace	Function
Overview	Project, dataset and model-performance summary.
Single-sample analysis	Entry of 30 WDBC measurements and model inference.
Batch laboratory	CSV upload, schema/numeric validation, batch predictions and optional label comparison.
Explainability	Displays saved coefficient and permutation-importance outputs.
Model & methodology	Pipeline, threshold, metrics and limitations.

The single-sample page uses reference-dataset medians as demonstration defaults and identifies them as demo values, not patient measurements. Metrics from user-uploaded labelled CSV files are exploratory and distinct from the recorded holdout metrics.

7. Evaluation Limitations and Risk Controls

Holdout-selection limitation: the notebook compares tuned Logistic Regression and SVM on the holdout test split and then selects Logistic Regression. The holdout therefore influenced model selection. Reported metrics are exploratory estimates, not an untouched final generalisation estimate. A new untouched test set should be used after model and threshold decisions are locked.

Clinical limitation: the project does not establish clinical validity, prospective performance, population calibration or suitability for patient care. External validation and qualified clinical review would be required before any clinical use could be considered.

Input limitation: predictions require the expected WDBC feature definitions, units and order. A model output probability is not a patient-specific medical risk estimate.

8. Reproducibility and Deliverables

Deliverable	Repository path
Streamlit application	app/streamlit_app.py
Data validation	src/validate_data.py
EDA / model notebooks	notebooks/01_data_exploration.ipynb; notebooks/02_model_development.ipynb
Saved model / metadata	models/breast_cancer_logistic_regression.joblib; models/model_metadata.json
Analysis artifacts	outputs/eda/; outputs/model_evaluation/
Dependencies / documentation	requirements.txt; requirements-dev.txt; README.md

Run locally: clone the repository, create and activate a Python virtual environment, run **pip install -r requirements.txt**, then launch **streamlit run app/streamlit_app.py**. Full setup instructions are in README.md.

9. Conclusion

ONCOLENS implements an end-to-end educational WDBC classification workflow: data validation, EDA, comparison of five classifiers, tuning of two candidates, a saved Logistic Regression pipeline, feature-importance artifacts and a hosted Streamlit prototype. The recorded results are strong on the selected split, but the holdout-selection limitation means a new untouched evaluation is needed before making stronger generalisation claims.

Project links

GitHub repository — Breast-Cancer-Detection-ML | Live ONCOLENS application

Report prepared from the repository README, model metadata, validation source and inspected model-development results. No new model training, independent validation or clinical testing is claimed.